Lab Tasks

1. Design a simple Client Server chat application which sends files using the above concepts.

2. Develop a chat server application that can handle multiple clients at once, allowing them to communicate in a chat room-style application. [HINT: Use Threading]

3. Modify the multi-client chat application (developed in Task 2) in Python using sockets to include server-side validation for messages and file transfers. Your application should meet the following requirements:
   - The server should only allow certain file types to be sent by clients (e.g., .txt, .jpg, .pdf). Files with disallowed extensions such as .exe should be rejected.
   - The server should scan text messages for some banned words or inappropriate content and reject or warn the sender if such content is detected.
   - Normal messages and allowed files should continue to be delivered to other clients as usual.
   - The server should handle multiple clients simultaneously without crashing.
   - Clients should receive appropriate feedback messages when their messages or files are rejected.

[Hint 1: Use the filename extension and string search methods to implement these validations.]
[Hint 2: Import Threading & Import os]